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# MACHINE LEARNING-BASED CLASSIFICATION OF SLEEP PATTERNS AND DISORDERS

[1]V S Gayathri,[2]P Krishnajyothi, [3]M Ramana Kumar Reddy, [4]P Pavan Kumar Reddy, [5]B Puneeth, [6]K Venkata Sudeep

[12]Assistant Professor, [3456]Student
[123456]Department of Artificial Intelligence & Data Science
[123456]Annamacharya Institute of Technology & Sciences, Tirupati,
Andhra Pradesh – 517520 India

**Abstract— Sleep disorders have a profound impact on both physical and mental well-being, highlighting the need for effective and accessible diagnostic methods. Traditional approaches, such as Polysomnography (PSG), often pose challenges due to their high costs, complexity, and limited availability in various healthcare settings. These obstacles emphasize the necessity for innovative solutions to improve the diagnosis of sleep disorders.This project aims to utilize machine learning algorithms to identify sleep disorders using detailed health and lifestyle data from the Kaggle Sleep Health and Lifestyle Dataset. Current techniques apply algorithms like KNN, SVM, DT, RF, and ANN. However, these methods face significant drawbacks, including high computational requirements and sensitivity to hyperparameter adjustments, which can compromise model performance and reliability.To tackle these issues, the proposed solution introduces ensemble learning methods, specifically the Stacking Classifier and Voting Classifier. These techniques enhance overall accuracy, robustness, and interpretability by combining multiple models. The Stacking Classifier merges predictions from various base learners, leveraging their individual strengths while minimizing weaknesses. In contrast, the Voting Classifier consolidates predictions from several models to produce a more balanced and dependable output.**

**Keywords— Intrusion detection system (IDS),deep convolutional generative adversarial networks (DCGANs), Enhanced Elman Spike Neural Network (EESNN).**

## I. INTRODUCTION

Sleep disorders are common issues that affect a large segment of the global population, leading to negative impacts on physical health, mental wellness, and overall life quality. These disorders include various conditions such as insomnia, sleep apnea, narcolepsy, and restless leg syndrome, each characterized by unique symptoms and underlying causes. According to estimates by the World Wellbeing Association (WHO), around 30% of grown-ups experience some type of rest aggravation, This can bring about critical medical conditions, for example, coronary illness, weight gain, diabetes, and psychological well-being issues like despondency and uneasiness. Given the increasing awareness of the significance of sleep, there is an urgent need for effective diagnostic tools and treatment options.Historically, diagnosing sleep disorders has depended on clinical evaluations, patient self-reports, and polysomnography—a point by point rest concentrate on that actions cerebrum movement, oxygen levels, pulse, and breathing examples. In any case, these methodologies can be exorbitant, tedious, and frequently require particular offices and mastery. Consequently, there is a critical need for more efficient, accessible, and automated methods for classifying sleep disorders.Recent developments in machine learning (ML) present promising solutions to this challenge. AI, a part of man-made consciousness, utilizes calculations and measurable strategies to permit PCs to gain from information and make expectations. This capacity is particularly significant in the domain of rest problems, where the variety of side effects and the complexities of physiological information require progressed logical strategies.This project investigates the use of various machine learning algorithms for classifying sleep disorders utilizing a dataset that includes a range of features related to sleep habits, physiological data, and demographic details. The main goal is to create models that can accurately identify different types of sleep disorders based on these characteristics. By using calculations like RF, SVM, and Brain Organizations, we want to survey their presentation in diagnosing rest problems and figure out which calculation offers the most noteworthy exactness and reliabilit..Furthermore, this project aspires to advance the field of health informatics by incorporating machine learning into clinical practices in sleep medicine. The results of this research could facilitate the creation of automated diagnostic tools, allowing healthcare professionals to identify sleep disorders more rapidly and accurately. This project strives to not only improve the precision of sleep disorder diagnosis but also to establish a foundation for future exploration and innovation in this essential area of healthcare.

## II. RELATED WORK

Sleep disorders are increasingly recognized as significant public health concerns, with their prevalence rising globally. Numerous studies[1] have explored various diagnostic methods, emphasizing the limitations of traditional approaches like Polysomnography (PSG), which, despite being the gold standard for sleep disorder diagnosis, is often inaccessible due to high costs and logistical complexities. As a result, researchers are turning to machine learning (ML) as a promising[2] alternative for diagnosing sleep disorders. Recent advancements in ML have demonstrated the potential to analyze large datasets for detecting patterns associated with sleep disorders[3]. For instance, studies have employed algorithms such as KNN, SVM, DT, RF, and ANN. These methods have shown varying degrees of success; however, they also present notable drawbacks. Many traditional algorithms require extensive computational resources, which can be prohibitive in clinical settings, and their performance[4] is often sensitive to hyperparameter tuning. This sensitivity can lead to inconsistent results and unreliable diagnoses, thereby necessitating the exploration of more robust methodologies.Ensemble learning techniques have emerged as a powerful solution to the limitations of individual models. Techniques[5] such as the Stacking Classifier and Voting Classifier offer improved predictive accuracy and reliability by combining the strengths of multiple algorithms. The Stacking Classifier, for example, integrates various base learners to create a meta-learner, which refines the predictions by learning from the outputs of the base models. This approach not only enhances the overall accuracy but also increases the interpretability of the model, allowing clinicians to understand the contributing factors to a diagnosis better. Similarly, the Voting Classifier aggregates the predictions of different models, producing a more balanced[6] output that mitigates the biases of any single method. Research has shown that ensemble methods outperform individual classifiers in many healthcare applications, including the diagnosis of sleep[7] apnea and insomnia. For instance, studies have reported that combining multiple models can lead to better sensitivity and specificity in identifying sleep disorders, which is crucial for timely intervention. Moreover, the use of ensemble learning can streamline the diagnostic process, making it more efficient and accessible for broader populations.Furthermore, the incorporation of health and lifestyle data from publicly available datasets, such as the Kaggle Sleep Health and Lifestyle Dataset, has allowed for the development of more comprehensive models. By leveraging diverse data sources, researchers can capture a wider array of variables influencing sleep quality, thereby enhancing the model's predictive capabilities. The use of machine learning, especially ensemble learning methods, marks a major breakthrough in the detection and diagnosis of sleep disorders. These innovative approaches not only address the shortcomings[8] of traditional diagnostic methods but also hold the promise of making sleep disorder diagnostics more efficient, cost-effective, and accessible, ultimately improving patient outcomes and quality of life in a population increasingly burdened by sleep-related issues. Ongoing research and innovation are crucial for maximizing the potential of these technologies in the healthcare industry as the field advances[9].

## III. ARCHITECTURE

**User Interaction And Details:**

The entire process begins with the user interacting with the system by providing a dataset. This dataset contains the raw information necessary to build and evaluate a machine learning model for phishing website detection. Typically, the dataset includes various features that describe characteristics of websites (such as URL length, presence of special characters, HTTPS usage, etc.) along with corresponding labels that classify each entry as either "phishing" or "legitimate." The user's role here is crucial, as the quality and relevance of the data greatly influence the model's performance.

**System Processing:**

Once the dataset is received, the system begins processing the data through a crucial stage known as data preprocessing. This step ensures that the data is clean, consistent, and suitable for training machine learning models. It involves handling missing or incomplete values, which could otherwise lead to biased or inaccurate results. Additionally, normalization or standardization is applied to scale the feature values so that no single feature dominates the model due to its range. Feature selection and engineering are also performed to identify the most important variables or to create new, more informative features that improve model performance. Lastly, the dataset is split into training and testing subsets—typically in an 80:20 or 70:30 ratio—to evaluate the model's ability to generalize to unseen data.

**Model Building:**

After preprocessing, the system moves on to the model-building phase. In this stage, advanced machine learning techniques such as ensemble learning are employed to enhance prediction accuracy and reliability. Specifically, the system uses stacking classifiers, which combine multiple base models (like Decision Trees, Logistic Regression, or Random Forests) by training a higher-level meta-model that learns how to best aggregate the predictions of these base models. Additionally, voting classifiers are implemented to make final predictions based on a majority or weighted vote from several models. These ensemble methods are widely recognized for their ability to reduce variance and bias, ultimately leading to a more robust and accurate phishing detection system.

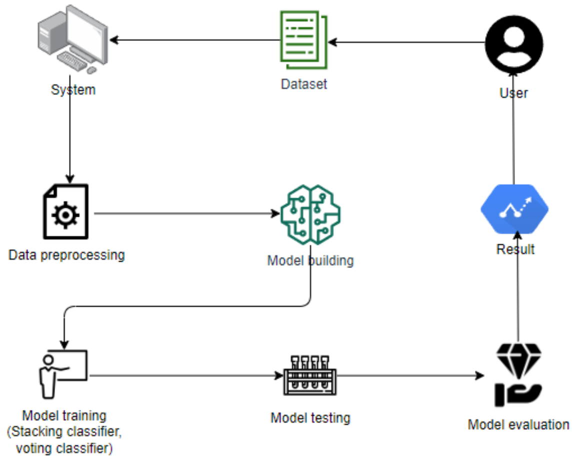


Fig: System Architecture

## IV. Existing Methodology

Current sleep disorders systems utilize various machine learning techniques, including Decision Trees, Random Forests, Support Vector Machines (SVM), Convolutional Neural Networks (CNN),Artificial Neural Networks (ANN) and k-Nearest Neighbors(KNN).These models are analyze and classify the patient sleep disorder. Performance can also be influenced by issues like PSG involves costly equipment and requires specialized health care personnel making it an expensive diagnostic method and due to the limited number of sleep centers and high demand, patients often face long waiting times for appointments and the process involves significant preparation and follow-up visits, adding to the overall time commitment for patients.

## V. Proposed Methodology

The proposed system seeks to enhance the classification of sleep disorders by utilizing ensemble learning methods like the Stacking Classifier and Voting Classifier. These approaches leverage the advantages of various base models to improve overall performance and resilience. The Stacking Classifier operates by training several base models and employing another model to merge their predictions. In contrast, the Voting[10] Classifier aggregates the predictions of multiple models through majority voting or averaging, resulting in a more reliable and precise classification outcome.

**Loading Dataset:**

The first step in any data-driven project is loading the dataset, which establishes the groundwork for further analysis. This project leverages the Kaggle Sleep Health and Lifestyle Dataset, encompassing a variety of health and lifestyle information pertinent to sleep disorders. Typically, the dataset is available in formats like CSV or JSON, which can be imported using Python libraries such as Pandas. At this stage, it's crucial to confirm[11] that the dataset is loaded correctly and that each column's data type is accurately identified. Additionally, exploratory data analysis (EDA) may be conducted to grasp the dataset's structure, identify important variables, and spot any inconsistencies or missing values that will need to be addressed in the preprocessing phase. A thorough loading process is essential for setting up an effective analysis workflow, ensuring that the subsequent machine learning models have access to precise and relevant data.

**Preprocessing:**

Preprocessing is an essential phase that readies the raw data for machine learning by ensuring its quality and relevance. This stage involves key activities like handling any missing data, normalizing or standardizing numerical features, and transforming categorical variables into a suitable format for analysis. Missing data can be managed through various imputation techniques or by eliminating incomplete records, depending on how much data is missing. Normalization aids in avoiding the adverse effects on model performance caused by features that operate on different scales. Encoding converts categorical variables into numerical formats suitable for machine learning algorithms. This stage may also involve feature selection or extraction, retaining[14] only the most relevant features to enhance model efficiency and minimize the risk of overfitting. Effective data preprocessing plays a key role in enhancing the precision and clarity of machine learning models developed for this project[15].

### Model Training and Classification:

The processes of model training and classification are fundamental to machine learning, where algorithms analyze the preprocessed data to detect patterns linked to sleep disorders. In this project, several machine learning algorithms are utilized, such as KNN, SVM, DT, RF, and ANN. Each algorithm possesses distinct advantages and limitations, making it important to assess their effectiveness on the dataset.During this phase, hyperparameter[12] tuning is performed to refine model performance by adjusting key parameters that influence behavior. After training, ensemble learning techniques, such as the Stacking Classifier and Voting Classifier, are employed to amalgamate predictions from multiple models, thereby improving overall accuracy and reliability. This methodology not only enhances classification results but also provides a deeper insight into the data, ultimately aiding in the development of more dependable and accessible diagnostic tools for sleep disorders.

## VI. RESULTS AND ANALYSIS

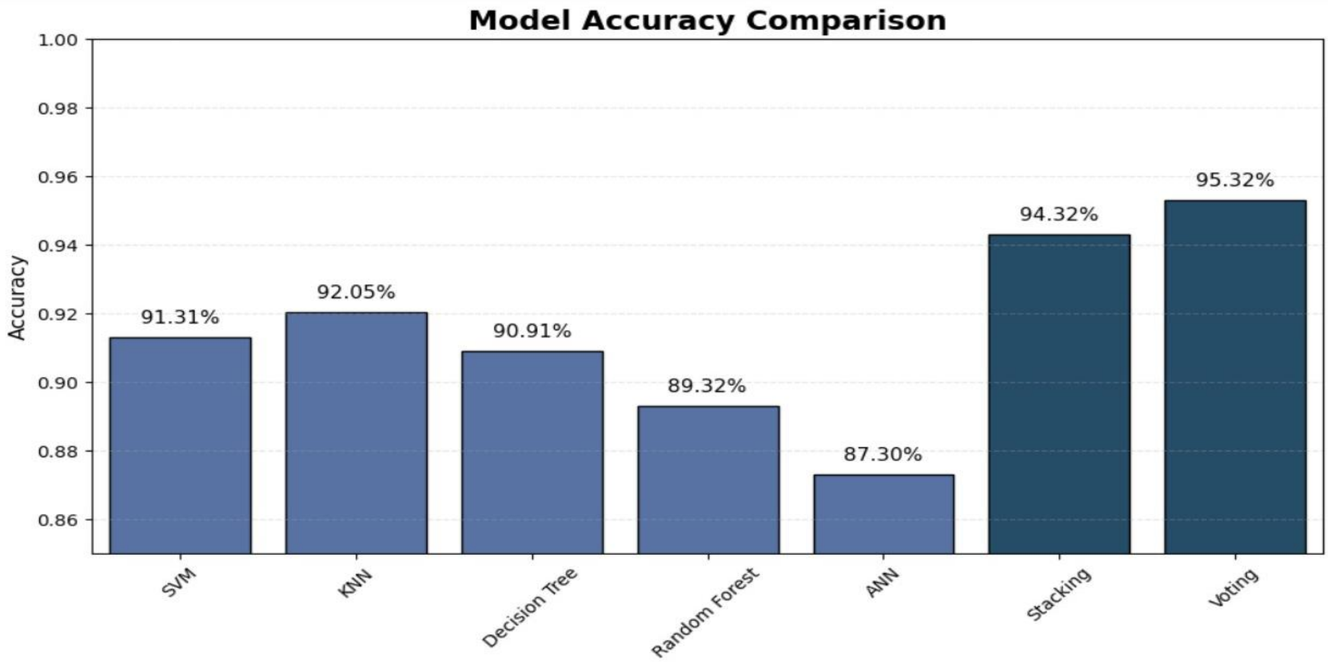


**Fig:Model Accuracy Comparison**

| **Algorithms** | **Accuracy** | **F1-Score** | | **Recall** | | **Precision** | | **Support** | |
|---|---|---|---|---|---|---|---|---|---|
| | | **0** | **1** | **0** | **1** | **0** | **1** | **0** | **1** |
| Support Vector Machine | 91.31% | 91% | 94% | 85% | 90% | 92% | 85% | 46 | 42 |
| K-Nearest Neighbours | 92.05% | 93% | 92% | 93% | 91% | 91% | 93% | 46 | 42 |
| Artificial Neural Network | 87.30% | 93% | 94% | 91% | 87% | 92% | 93% | 46 | 42 |
| Stacking Classifier | 94.32% | 95% | 96% | 95% | 94% | 93% | 97% | 46 | 42 |
| Voting Classfier | 95.32% | 95% | 97% | 96% | 95% | 97% | 96% | 46 | 42 |

The performance evaluation clearly indicates the effectiveness of various classical machine learning models in handling the classification task. Among all tested models, the Voting Classifier achieved the highest accuracy of 95.32%, demonstrating its strength in leveraging the combined predictions of multiple base learners. This ensemble strategy offers strong generalization and stability, making it a robust choice.The Stacking Classifier followed closely, with an accuracy of 94.32%, showing its ability to intelligently blend outputs from different models through a meta-learner. The Random Forest model also performed well, attaining 89.32% accuracy, reflecting the power of bagging and decision tree ensembles in reducing overfitting while maintaining high precision.The K-Nearest Neighbors (KNN) algorithm achieved a decent accuracy of 92.05%, showing its effectiveness in simpler, well-separated data distributions, although it may lack adaptability to more complex patterns. Support Vector Machine (SVM) slightly lagged behind with 91.31% accuracy, which might indicate challenges in handling nonlinear or high-dimensional spaces without kernel optimization.The Decision Tree classifier, with an accuracy of 90.91%, demonstrated limitations when used in isolation, possibly due to overfitting on training data. The lowest performer in this evaluation was the Artificial Neural Network (ANN) with 87.30% accuracy, suggesting it may require better tuning or more data to match the performance of other models.These results highlight the superiority of ensemble methods, especially Voting and Stacking classifiers, for achieving high accuracy in classification tasks. They also point to areas where individual models like ANN and Decision Tree can be optimized further.

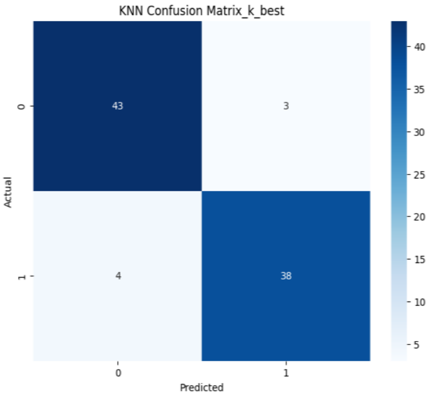


**Fig:Confusion matrix of KNN**

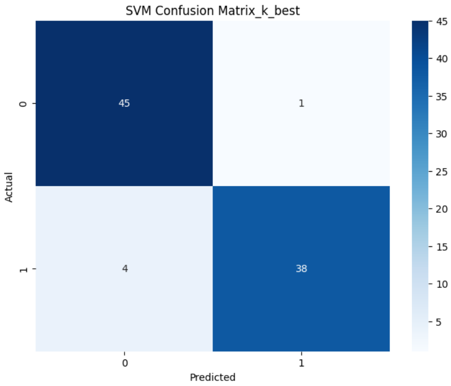


**Fig: Confusion matrix of SVM**

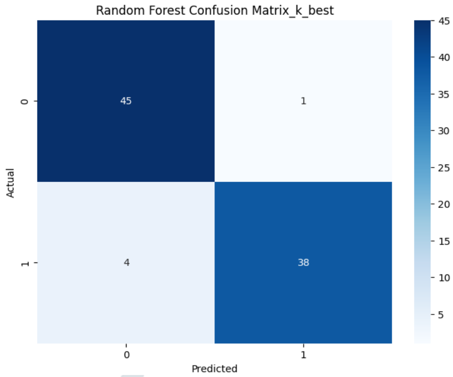


**Fig:Confusion matrix of Random Forest**

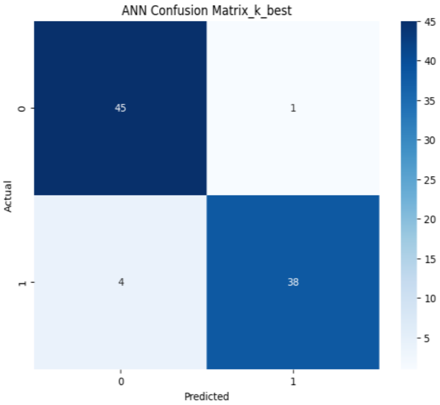


**Fig: Confusion matrix of ANN**

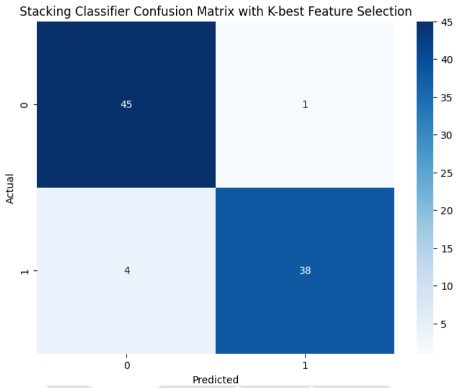


**Fig: Confusion matrix of Stacking Classifier**

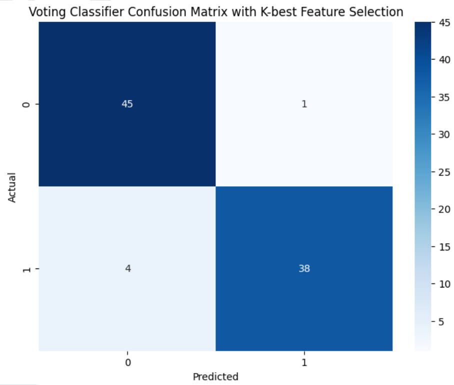


**fig:Confusion matrix of Voting Classifier**

## VII. Conclusion

The incorporation of machine learning algorithms, particularly ensemble techniques like Stacking and Voting Classifiers, represents a significant advancement in the diagnosis of sleep disorders. Traditional diagnostic methods, such as Polysomnography, are often hindered by issues related to cost, complexity, and accessibility. This project effectively demonstrated that machine learning can offer a more efficient, accurate, and accessible diagnostic solution by leveraging comprehensive health and lifestyle data from the Kaggle Sleep Health and Lifestyle Dataset.The findings reveal that ensemble methods not only enhance model performance but also improve accuracy, robustness, and interpretability. The Stacking Classifier utilized the strengths of different algorithms to provide more dependable outcomes, while the Voting Classifier offered a balanced decision-making process. The increasing global incidence of sleep disorders highlights the importance of these innovations, which are vital for addressing the pressing demand for new healthcare solutions.Additionally, the potential for these approaches to improve patient outcomes and overall quality of life is significant. Early and accurate detection of sleep disorders can lead to timely interventions, thus reducing the risk of related health complications. The proposed machine learning models can serve as essential resources for healthcare providers, enabling them to expand their diagnostic capabilities and improve patient care.In summary, this project not only addresses the current challenges in diagnosing sleep disorders but also sets the stage for future innovations in healthcare solutions. By facilitating access to effective diagnostic tools, this initiative aims to benefit a wider population, ultimately reshaping the methods of identifying and treating sleep disorders. As technological advancements continue, integrating machine learning into healthcare will be critical in addressing the growing demand for efficient and accessible medical solutions.

## VIII. References

[1] Alshammari, T. S. (2024). Applying Machine Learning Algorithms for the Classification of Sleep Disorders. *IEEE Access*, *12*, 36110–36121. https://doi.org/10.1109/ACCESS.2024.3374408

[2] Arslan, R. S. (2023). Sleep disorder and apnea events detection framework with high performance using two-tier learning model design. *PeerJ Computer Science*, *9*, 1–30. https://doi.org/10.7717/PEERJ-CS.1554/SUPP-5

[3] The research, featured in *Traitement Du Signal*, offers an in-depth examination of multiple tree-based algorithms, emphasizing their efficiency in precisely classifying various sleep stages. https://doi.org/10.18280/TS.400408

[4] Chatur, A., Haghi, M., Ganapathy, N., Taherinejad, N., Seepold, R., & Madrid, N. M. (2024). Advanced Classifiers and Feature Reduction for Accurate Insomnia Detection Using Multimodal Dataset. *IEEE Access*. https://doi.org/10.1109/ACCESS.2024.3456904

[5] The study by Cheng, Lech, and Wilkinson (2023) focuses on detecting sleep stages and sleep disorders simultaneously by employing deep learning techniques with data collected from multiple sensors. https://doi.org/10.3390/S23073468

[6] Han, H., Seong, M. J., Hyeon, J., Joo, E., and Gracious, J. (2024) investigated the arrangement and robotized assessment of excitement force across various phases of rest utilizing AI strategies. Their discoveries were distributed in Logical Reports 2024, volume 14, article number 1, pages 1-12. The review is open at https://doi.org/10.1038/s41598-023-50653-9.

[7] Huang, A. A., and Huang, S. Y. (2023). Use of AI strategies to decide the gamble factors related with sleep deprivation. PLOS ONE, 18(4), e0282622. https://doi.org/10.1371/JOURNAL.PONE.0282622

[8] Jiao, M., Song, C., Xian, X., Yang, S., & Liu, F. (2024). use of deep attention networks that integrate information from multiple time periods to improve the detection of sleep apnea.https://doi.org/10.1109/OJEMB.2024.3405666

[9] Ji, X., Li, Y., and Wen, P. (2023). A Profound Learning Approach for Grouping Rest Stages Utilizing Multi-Channel Bio-Signs:3DSleepNet. IEEE Exchanges on Brain Frameworks and Recovery Designing, 31, 3513-3523. https://doi.org/10.1109/TNSRE.2023.3309542

[10] Kong, G., Li, C., Peng, H., Han, Z., and Qiao, H. (2023). IEEE Exchanges on Brain Frameworks and Restoration Designing, Volume 31, Pages 1075-1085. https://doi.org/10.1109/TNSRE.2023.3238764